Mal
SME Banking · UAE

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Companion

01 Visual architecture diagram

Document

v1.0 · 2026-05-13
Confidential, for Mal exec and board

MAL-SME-Banking

Architecture documentation. A reference architecture, a vendor stance, a team model, and a build sequence for launching Mal. The launch surface is four things: multi-currency accounts, Murabaha-compliant invoice financing, expense management, and integrated accounting. Around them sits an AI-native decisioning platform, a Sharia compliance engine, and a regulator-aligned audit fabric.

Customer

UAE SMEs of 5 to 50 employees, mainland and free zone.

Licensing path

ADGM FSP Category 2 plus Category 4, with optional RegLab sandbox.

Cloud

AWS me-central-1 in Abu Dhabi, DR in me-south-1.

Sharia

AAOIFI-aligned, SSB-governed, audited quarterly.

Executive summary

Mal is an SME bank for the UAE, serving businesses of 5 to 50 employees. The launch surface is four things. Multi-currency accounts. Invoice financing structured as Murabaha. Expense management. Integrated accounting with native connectors for the tools UAE SMEs already use. Around those four sits an AI-native decisioning platform, a Sharia compliance engine, and a regulator-aligned audit fabric.

The principles

Build over buy. Software is cheaper to build than it has been at any point in our careers. With a small, senior, AI-native engineering team and modern tooling, the parts of the stack that touch our customers, our policy, or our IP are faster to build than to integrate, and we end up owning the product. We buy where the value is genuinely ecosystem-wide (fraud, sanctions), where the regulator mandates the counterparty (AECB), or where the system is a true commodity (cards, identity rails). Everything else is built in-house.

Sharia by design. Every product is contract-typed in the data model. Invoice finance is a true-sale Murabaha, cost plus profit, deferred payment, under a fatwa from Dar Al Sharia or Shariyah Review Bureau, AAOIFI-aligned, and submitted to the UAE Higher Sharia Authority before launch. Late fees never accrue to Mal income. They go to charity, and the customer sees the destination on their statement.

UAE-resident and CBUAE-aligned. The primary licence is ADGM FSP Category 2 (Providing Credit) plus Category 4 (Operating a Private Financing Platform), with ADGM RegLab as an optional sandbox for the BNPL instalment feature inside invoice finance. The CBUAE Customer Protection Regulation for SMEs (effective 13 Sep 2026) is designed into product disclosures from day one. All customer PII and ledger journal data lives on AWS me-central-1, with warm DR in me-south-1. Audit trails are immutable for ten years.

§ 02 · Principles

Architectural principles

Eight tenets that decide every design trade-off. When a vendor pitches, when a team debates monolith versus services, when the SSB asks for a contract change, these are the rules we go back to.

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1 · Build over buy, with a tight AI-native team

Software is cheaper to build than at any point in our careers. We staff small, senior, AI-native engineering pods and let them own the parts of the stack that touch the customer, the policy, or the IP. We buy only where the value is genuinely ecosystem-wide, where the regulator mandates the counterparty, or where the system is pure commodity.

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2 · API-first, headless core

The product surface is one of many clients. Every domain capability is a versioned API before it is a UI. That is what lets us ship an iOS app, a web app, and a partner API on the same primitives.

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3 · Event-sourced ledger

The ledger is the source of truth, and the ledger is a stream. Balances are a projection. Statements are a projection. Analytics are a projection. We can rebuild any view of money by replaying journal events.

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4 · Cloud-native, multi-AZ, UAE-resident

Everything stateful runs on managed AWS services in me-central-1, replicated across three AZs. Nothing customer-identifying leaves the UAE without explicit consent and a CBUAE notification.

5 · Sharia by design

Contract type is a first-class field on every money-moving record. The system refuses to construct a transaction that has not been approved in template by the SSB.

6 · Zero-trust by default

No service trusts another by network position. mTLS between services, OIDC for users, signed receipts on every state transition. Step-up auth for risk events.

7 · Decisioning is a mix, not a model

Underwriting, fraud, EWS, pricing, collections. Each is some combination of deterministic rules, written policies, manual workflows, referrals, and models. The model class varies by problem. We do not commit to one algorithm family across the platform.

8 · Audit-first

If it cannot be audited, we do not ship it. Every record carries provenance: who decided, when, on what model version, with what input snapshot. Every AI decision logs SHAP contributions for the regulator and a plain-language reason for the customer.

Core banking infrastructure

Three sub-systems: account management (who the customer is, what wallets they hold), transaction processing (how money moves in and out), and the ledger (the canonical record of every cent).

3.1 · Account management

The LMS is licensed from Mambu Cloud on day one (the same core Wio Bank runs in the UAE), and the plan is to migrate to an in-house LMS once we understand our exact requirements at scale. Licensing first means we ship without inventing a deposit and loan engine from scratch; migrating later means we keep the option open to specialise the ledger for Sharia-typed contracts and to control the cost curve.

Around the LMS we build:

- **Customer service**. Holds the canonical KYB record: trade licence (DET, DED, ADGM, DIFC live pull), UBO graph (≥25% per UBO per CBUAE Rulebook §3.1, ≥80% aggregate coverage), authorised signatories, employees with delegated authority. The graph is versioned and every change generates an event. UBO pull is automatic via the Ministry of Economy register.
- **Multi-currency accounts**. AED is primary. USD, EUR, GBP are sub-ledgers held with Currencycloud as the FX counterparty. Virtual IBANs are issued through a partner bank (Wio or RAKBank under a BIN-sponsorship arrangement).
- **Expense management and cards**. Virtual and physical issuance via Nymcard. Per-employee spend controls (MCC blocks, daily and monthly caps, single-merchant locks). 3DS via Visa Token Service. Receipt OCR for Arabic-English VAT invoices auto-categorises spend, and a policy engine enforces approval workflows.
- **Integrated accounting**. Native connectors for Wafeq (UAE-native, Peppol-ready), Zoho Books, QuickBooks Online, Xero, and Tally Prime; Codat as the meta-integrator for the long tail. Every Mal transaction posts to the customer's GL within 60 seconds.
- **Consent and delegation**. Explicit, time-bound, revocable. Every API call carries the consent scope it was issued under, validated at the gateway.

Three delivery variants for invoice finance

Variant	*Ticket*	*Touch*	*SLA*	*Inputs*
Express	≤ AED 250K	Fully automated	< 4 hours	On-us 12-month data only
Standard	AED 250K – 750K	Semi-automated · AI + human	24–72 hours	+ VAT returns + 6-mo off-us bank statements
Enhanced	AED 750K – 2M	RM-led hybrid	3–5 days	+ audited financials + RM call + sector model

3.2 · Transaction processing

A single Payments Orchestrator service decides which rail to use, holds idempotency keys, manages retries, and runs the compensating saga if downstream fails. It writes to the ledger first (pending), then sends to the rail; on confirmation it posts the final journal; on rail failure it issues a reversal.

```
// Payments Orchestrator · pseudocode for an outgoing AED transfer
async function sendAED(req) {
  const idem = req.idempotencyKey;
  const pending = await ledger.writePending({ idem, dr: customer, cr: suspense, amou
  const rail   = selectRail(req); // aani | uaefts | swift
  try {
    const ack  = await rail.send({ idem, ...req });
    await ledger.commit({ idem, dr: suspense, cr: counterparty, ref: ack });
  } catch (err) {
    await ledger.reverse({ idem });
    throw err;
  }
}
```

Rail selection rules:

- AED domestic ≤ AED 1M and beneficiary at a connected bank → **Aani** instant.
- AED domestic > AED 1M, or beneficiary at a non-Aani-connected bank → **UAEFTS** RTGS.
- Cross-border GCC AED corridor → **Buna**.
- Everywhere else → **SWIFT** via partner bank.

3.3 · The ledger

The ledger is double-entry, append-only, and event-sourced. Every transaction produces one or more journal entries; each entry is a Kafka event with a strict Avro schema. Balances are a materialised projection refreshed in real-time on the OLTP side and lazily on the analytics side.

Each journal record carries:

- **Contract type**. Murabaha for invoice finance, plus the typed contract for any other product. This is what makes Sharia review tractable at scale.
- **Tenant chain**. The source SME, the counterparty, the product, the underlying invoice or asset.
- **Provenance**. Service, version, decision model, input hash.
- **Reversibility window**. Until what timestamp can this entry be reversed without dual approval.

> *Why we license the LMS at launch and migrate later*
>
> *Ledger accuracy is non-negotiable, and getting it wrong on day one is not something we want to debug under production pressure. Mambu gives us deposit and loan products, regulator-friendly defaults, and integrations that already pass UAE bank audit. We license at launch and build the in-house replacement once we know exact requirements from live operations. The orchestration around it (Payments Orchestrator, Murabaha contract engine, Customer graph) is ours from day one, because that is where the product lives.*

Third-party integrations

Every external dependency is fronted by an adapter service that owns retries, schema mapping, rate-limit budgets, and circuit-breaking. The domain services never call a vendor SDK directly.

4.1 · Payment rails

Rail	*Owner*	*Use case*	*Latency*	*Notes*
Aani	CBUAE / Al Etihad Payments	AED instant retail + SME, account-to-account	< 10s	Mandated rail for instant; SDK via partner bank
UAEFTS	CBUAE	AED RTGS for high-value	Same day	For ≥ AED 1M or non-Aani-connected counterparties
IPP / IPI	CBUAE	Interbank file transfers (declining)	Batch	Mostly legacy; Aani is replacing
Buna	Arab Monetary Fund	Cross-border AED + regional	Hours	GCC corridor; growing
SWIFT	SWIFT (via partner)	USD / EUR / GBP cross-border	0–2 days	Through correspondent at Wio or RAKBank
Visa / Mastercard	Schemes (via Nymcard)	SME spend cards, virtual + physical	Authorise: 200ms	3DS + Visa Token Service for tokenisation

4.2 · KYC / KYB / AML

Three layers, run in parallel during onboarding and continuously thereafter (perpetual KYB).

– **Identity** · UAE Pass federated; Emirates ID OCR + face-match via Sumsub as fallback for non-UAE-Pass principals. Sumsub FOCAL

- **Business** · trade licence + MoA + UBO graph; live DED/JAFZA/ADGM register lookups; Sumsub KYB module. `Sumsub KYB`
- **Screening** · sanctions, PEP, adverse media, in real-time at first onboarding and then continuously. Hits flow to the Compliance team as alerts. `ComplyAdvantage` `Refinitiv World-Check`

The KYC orchestration is a Temporal workflow with explicit states (`submitted → screening → manual_review → approved | rejected`) and SLA timers; if any stage hangs past its budget the workflow pages the on-call AML analyst. Outcomes write to the customer-graph event stream so downstream services see consistent state.

4.3 · Credit scoring & alternative data

Two sources for every credit decision:

- **AECB** · the UAE credit bureau. Mandatory pull for any SME credit product. Cached for 90 days; refreshed on material event (new application, default, bureau notice).
- **Alternative data** · 12 months of bank statements via Lean (read-only open banking), accounting data via Wafeq/Zoho/QuickBooks connectors, e-invoicing data via Peppol, payment-rail history (if the SME holds a Mal account). These feed the cash-flow forecasting and underwriting models.

4.4 · Accounting software

The integrated accounting layer is what turns Mal from a bank into a finance operating system. We support four UAE-relevant tools natively, and many more via Codat (a meta-integrator) as a fallback:

- **Wafeq** · UAE-native, Peppol-ready, FTA-aligned. Default recommended tool for new SMEs onboarding.
- **Zoho Books** · Strong UAE penetration; API-first.
- **QuickBooks Online / Xero** · For SMEs with international owners or auditors.
- **Tally Prime** · For SMEs with Indian ownership / supply chain.
- **Codat** · Meta-integrator covering ~30 long-tail tools and on-prem (Sage 50, Microsoft Dynamics) at the cost of higher latency.

4.5 · Open banking aggregation

Lean Technologies is the primary UAE open-banking aggregator. We use it to pull 12 months of bank statements from the customer's existing bank (Emirates NBD, ADCB, Mashreq, FAB, ADIB, Wio) for underwriting and cash-flow modelling. **Tarabut Gateway** is the backup for the broader MENA footprint.

4.6 · FX & multi-currency

Currencycloud is the FX engine and multi-currency holding partner. Mal holds USD, EUR, GBP balances with Currencycloud sub-ledgers; the Treasury team books spot and forward contracts daily to hedge net positions. AED stays on the partner bank's books.

Sharia compliance layer

Five integrated mechanisms: governance, contract engine, real-time screening, charitable purification, and audit trail. AAOIFI standards are the baseline; CBUAE's Higher Sharia Authority circulars are the local overlay.

5.1 · Governance and the Sharia Supervisory Board

Three external scholars (AAOIFI-certified, at least one with UAE Higher Sharia Authority recognition) and an internal Head of Sharia who is full-time staff. The SSB meets monthly. Its outputs:

- **Standing fatwa** for the Murabaha invoice-finance template, refreshed quarterly and at every material change.
- **Contract templates** · text and structure pre-approved; the engine refuses to render a contract outside template.
- **Audit sample** · 5% of executed Murabaha contracts each quarter, drawn randomly, reviewed for structural conformity.
- **Annual public report** · published on Mal's website, addressed to customers.

5.2 · The Murabaha engine

Invoice financing's mechanic: Mal buys the receivable from the supplier (acquires an asset), then sells the underlying obligation to the buyer at a disclosed cost-plus markup with deferred settlement. Mal's profit is the markup, not interest on a loan. The system enforces this by structure: there is no "loan" entity for invoice finance; there is a `murabaha_contract` with an asset, a markup, and an EMI schedule.

What the engine enforces at runtime:

- Asset existence. There is a real, deliverable good or service backing the invoice.
- Pre-purchase. Mal must own the asset (briefly) before selling it to the buyer. The time gap is logged.
- Disclosure. The buyer sees Mal's cost, profit margin, and tenor before signing the binding promise (*wa'd*).
- No re-pricing. The markup is fixed at signing. Roll-overs are new contracts, not amendments.

5.3 · Halal screening

Every counterparty (supplier or buyer) is screened at onboarding and at every transaction. We block industries deemed haram by AAOIFI: alcohol, gambling, conventional financial services, adult entertainment, pork, weapons, tobacco. Industry classification uses NAICS / ISIC mapped to AAOIFI's exclusions; **IdealRatings** provides the ongoing reference data.

5.4 · Late fees → charity

If an SME pays late, a small penalty is charged, but it cannot accrue to Mal income. Penalty amounts are diverted into a registered UAE charity (for example Emirates Red Crescent), and the SME sees the destination on their statement. This is a structural choice, not an opt-in. The ledger refuses to credit penalty income to Mal's P&L accounts.

5.5 · Zakat calculation

For Sharia-electing SMEs, Mal calculates the annual Zakat due on net business assets. The tool tracks *hawl* (one lunar year ownership) per asset class, applies the 2.5% rate on the net *nisab*-eligible amount, and surfaces the calculation in the customer's home dashboard. Mal does not collect or remit Zakat; the SME pays directly to the charity of their choice.

5.6 · The audit trail

Every Sharia-relevant decision (template approval, screening result, contract execution, late-fee diversion) writes an immutable event. The SSB's quarterly audit pulls a stratified random sample directly from the event store; the auditor's read-only console queries the same data the engine sees.

§ 06 · AI / ML

AI / ML components

Decisioning at Mal is a mix, not a model. Every decision we make about a customer, whether it's underwriting, pricing, EWS, fraud, collections, or pre-approval, runs through the same platform. The platform composes deterministic rules, written policies, manual workflows, referrals, and models. The model class varies by problem. The platform stays the same.

6.1 · The five-layer AI architecture

This mirrors how the AI section is laid out in the Mal prototype, so that the architecture and the working app stay in sync.

Layer	*What it contains*
1 · Customer surfaces	Buyer app, supplier app, anchor portal, ops console. Every surface routes intent through the same platform.
2 · Agent orchestration	Tool router, replay log, eval harness, human-override. Deterministic state machinery (Temporal + LangGraph) that calls model-backed agents as steps. Not an LLM itself.
3 · Agent fleet	~20 specialist agents grouped by lifecycle stage (origination, underwriting, servicing, ops, risk and CX). Each agent has one job, one toolset, a golden eval set, and a human-in-the-loop trigger.
4 · Models and features	A feature store that holds the canonical inputs (cashflow, bureau, exposure, sector, behaviour). A model registry that holds the model class best suited for each task. An embedding store for retrieval. A small set of LLMs routed by job.
5 · Data foundations	AECB, EmaraTax e-invoice, UAE Open Finance, POS feeds (Network + Magnati), internal events. Cleaned, versioned, lineage-tracked.

A loan request descends from a customer surface through the orchestrator, fans out to the agents the request graph requires, reaches into the models and the feature store and the bureau and e-invoice and Open-Finance pulls, and returns with an explainable decision, a price, and a digital contract. Every action is hashed into the audit ledger.

6.2 · The Decision Engine, in five steps

This is the operational heart of the platform. It runs the same way for every decision the bank makes about a customer, whether the answer is "approve and price", "decline with a reason", or "escalate to a human". A risk manager can pause at any step, replay the trace, override a feature, and re-run.

Step	*What it does*
1 · Data pull	Bureau, e-invoice match, bank feed, KYC docs, POS data, prior platform footprint. Pulled in parallel where possible; cached behind the feature store.
2 · Feature build	The feature store builds the canonical inputs the model and rule layers need. Cashflow ratios, exposure, sector, seasonality, behaviour. Versioned, with lineage.
3 · Score	One or more models run, depending on the question. The class of model is chosen for the task: behaviour and deep-learning approaches in acquisition, gradient-boosted trees or survival models in underwriting and EWS, simpler classifiers in routing. The output of this step is a probabilistic signal, not a decision.
4 · Policy gate	Deterministic rules. Sharia, sanctions, exposure caps, DSCR floors, KYC pass/fail. The gate is binary and auditable. It overrides the model in either direction.
5 · Price and decide	Risk-tier pricing, limit, repayment plan ladder, customer-facing reasons. SHAP attribution for the regulator. Plain language for the customer.

> *This same engine runs for everything*
>
> *Acquisition (who do we proactively target?), onboarding (do we approve this KYB?), underwriting (limit and price), in-life servicing (limit review, EWS), collections (intent-to-pay routing), fraud (real-time scoring). Same plumbing, different mix of rules, policies, manual workflows, referrals, and model classes. There is one engine, not seven.*

6.3 · The 20-agent fleet, by lifecycle stage

Each agent owns one job. The grouping mirrors the customer lifecycle, so a risk owner or an ops lead can find every agent that touches their stage in one place.

Stage	*Agents*
Origination & onboarding	KYC Verifier, AECB Bureau Puller, Document Extractor, Cashflow Analyser
Underwriting & decisioning	Fraud Sentinel, Invoice Validator, Decision Engine, Pricing Optimiser
Servicing & collections	Schedule Builder, Dunning Orchestrator, Extension Recommender, AI Collections Voice (bilingual)
Operations & finance	Reconciliation Agent, Settlement Router, Audit Trail Composer, Anomaly Detector
Risk · compliance · CX	Sanctions Screener, Early-Warning Sentinel, Sharia Validator, Bilingual CX Co-pilot

6.4 · Where the model classes vary

We do not commit to one model family across the platform. The right model depends on the question.

Use case	*Model class*
Acquisition, propensity, look-alike, channel attribution	Behavioural and deep-learning approaches over transaction sequences and embedded representations of customer features
Underwriting (PD, LGD, affordability, limit sizing)	Gradient-boosted scoring families with monotonic constraints, SHAP attribution, and a parallel rules layer for adverse-action reasons
Early warning and hardship	Survival models for time-to-event, classifier ensembles for risk banding, change-point detection for behaviour shifts
Document understanding (OCR, layout, table extraction)	Layout-aware document models with ground-truth feedback loops
Identity, fraud, anomaly	Streaming feature pipelines with anomaly-detection baselines plus supervised classifiers ensembled on top
Conversational (CX, credit memo drafting, retrieval)	Retrieval-augmented LLM with structured-output enforcement and policy grounding. LLMs draft and explain. They never sign.
Optimisation (limit sizing, settlement routing, pricing)	Constrained optimisation with governance floors and caps; Bayesian search for combinatorial choices

Within each row, the specific algorithm choice is the team's. The architecture commits to the platform that lets them choose, version, evaluate, and replace algorithms on demand. It does not commit to one algorithm.

6.5 · Rules, AI, and the manual escape hatch

Layer	*What it handles*
Deterministic rules	Sanctions, blacklist, minimum eligibility, exposure caps, DSCR floors, sector concentration, KYC pass/fail. Binary, auditable, regulator-defensible.
Policies	Risk-team-owned, configurable in the Decision Engine without engineering lift. Changes are versioned and approved by the Credit Forum.
AI	Probability of default, affordability, dynamic pricing, EWS, document verification, fraud, alt-data scoring, conversational support.
Manual review	Any case the AI scores below a confidence threshold, any ticket above the Express ceiling, any first-time pair above policy, any sector flagged by the SSB. A credit officer reviews the model-generated memo, modifies limit or price, or declines with documented reason.
Referrals	Out-of-policy cases route to a named committee with a clear SLA. Cross-product referrals go to the Credit Forum. Sharia structure questions route to the SSB. Nothing falls through.

6.6 · Latency tiers

Latency	*What runs here*
Real-time (sub-second)	Fraud at origination, sanctions, drawdown approval, repayment processing
Near-real-time (under 4 hours)	Express full application, dynamic limit booster, supplier wire
Asynchronous (batch)	Portfolio and EWS scoring, limit reviews, bureau refresh, model retraining

6.7 · Explainability and right to human review

Every AI decision carries SHAP feature contributions. The top three to five features are translated into customer-facing reason codes through a mapping layer. Decline messages are specific and actionable:

Sample decline message

At this time we are unable to approve a facility. The primary reasons are: your business has been operating for 9 months (we look for 12 or more), and a dishonoured cheque of AED 18,000 is reported on your bureau in August 2025. We will re-evaluate automatically in 3 months. You can also request a human review now.

The right to human review is mandatory under the CBUAE Customer Protection Regulation (effective 13 Sep 2026) and is visible on every decline screen, not buried in T&Cs. Declined customers also feed a Credit Builder programme. Each gets a specific, achievable improvement target, weekly progress in-app, and automatic re-evaluation at 90 days. In Year 2 to 3 we expect 15 to 25 percent of new credit customers to come from the Credit Builder pool, which is a structural advantage over fintechs with no operating-account base.

6.8 · Fail-safe and fallback

Component	*Failure scenario*	*Fallback*
Credit scoring model	Model server down or low confidence	Queue for next run (max 4 hours), rules-only simplified scoring
AECB API	Bureau unavailable	Retry three times over two hours, escalate to manual review
Open Finance	Bank not connected via Lean	Fall back to uploaded statements, secondary post-dated cheque
Fraud detection	Model unavailable	Conservative rules plus human review for any drawdown above AED 100K
Conversational AI (LLM)	Provider outage	Route to human, pre-canned FAQ flows, no service disruption

What we do not use LLMs for

Underwriting decisions, fraud denials, Sharia contract construction, or anything that creates a binding obligation. LLMs are excellent at explanation, retrieval, drafting, and routing. They are not the model of record for a regulated decision. Customer-facing AI is always disclosed, and opt-out to a human is always available within two clicks.

Regulatory & compliance

The CBUAE is the primary regulator. AECB, the FTA, and the UAE FIU (via goAML) are the operational counterparties Mal interacts with daily. ADGM and DIFC are alternative regimes for some specific entities.

7.1 · Licensing path

The realistic 24-month progression:

1. **Phase 0 (M1–M6) · ADGM FSP Category 2 (Providing Credit) + Category 4 (Operating a Private Financing Platform)**. Optionally apply for **ADGM RegLab** sandbox for the BNPL instalment feature inside invoice financing (12–18 month sandbox period). CBUAE compliance for cross-border payments via partner bank.
2. **Phase 1 (M6 to M12)**. CBUAE Stored Value Facility (SVF) supervision via partner bank, which enables Mal-branded multi-currency wallets and cards.
3. **Phase 2 (M18 to M24)**. Specialised Bank licence consideration for the SME segment, which enables deposit-taking and native IBAN issuance at scale.

The architecture is designed so the same code base serves all three regulatory contexts; the difference is in product configuration, capital treatment, and reporting cadence.

CBUAE Customer Protection Regulation (effective 13 Sep 2026)

Product disclosures, complaints handling, and cooling-off rights are designed around this regulation from day one. Specifically:

- Plain-language disclosure of total cost of credit (not just rate) at every draw.
- Specific, actionable decline reasons surfaced via SHAP → customer-facing reason codes.
- Right to human review prominently visible on every decline screen.
- No compound interest on overdue amounts (aligned with CBUAE Law 2025); Ta'widh-only on the Sharia path.
- Complaint escalation path to CBUAE within 30 days, surfaced in-app.

CBUAE Open Finance Regulation (Circular 3/2025, effective 10 July 2025)

API-based access to customer financial data via the Nebras API Hub. **Lean Technologies** is the first In-Principle Approval (IPA) holder and our primary aggregator; **Tarabut Gateway** is secondary. Consent is captured per-account, time-bound, and revocable; the consent register is itself part of the audit log.

Federal e-invoicing mandate

Pilot live July 2026; mandatory for large taxpayers Jan 2027; mandatory for SMEs July 2027. Mal integrates with two FTA-accredited ASPs (**Avalara** + **RTC Suite** primary; **Pagero** / **Comarch** / **Tradeshift** as fallback) to avoid single-point failure. Peppol PINT AE XML is the canonical invoice format from day one.

7.2 · Reporting obligations

Report	*Frequency*	*To*	*Source system*
Liquidity Coverage Ratio	Daily	CBUAE	Treasury data mart + Mambu
Capital Adequacy	Monthly	CBUAE	Finance + risk data marts
Credit positions (AECB)	Monthly	AECB	Lending data mart
Suspicious transaction reports (STR)	Real-time	UAE FIU via goAML	AML alerts queue
FATCA / CRS	Annual	FTA	Customer + tax residency data
VAT returns	Quarterly	FTA	Finance ledger
E-invoicing (Peppol)	Real-time	FTA via accredited ASP	Invoice + accounting service
SSB report	Quarterly + Annual	Internal SSB + public	Sharia compliance engine

7.3 · Audit trail & retention

Every regulated event (KYC outcome, credit decision, money movement, Sharia approval, customer consent change) writes to an append-only event stream with WORM-immutable storage on S3 Object Lock. Retention floors:

- KYC artefacts and CDD data: 10 years post-relationship end (CBUAE AML guidance).
- Transaction journals: 5 years (CBUAE Banking Law).
- Audit logs (system access, admin actions): 7 years.
- Sharia decisions and contracts: 10 years.

The audit log is queryable through a read-only console used by Internal Audit, Regulator-Facing Compliance, and the Sharia Supervisory Board. External auditors get a time-boxed read-only role.

Security & data residency

Defence-in-depth, with the explicit assumption that any single layer can fail.

Data residency

Customer PII, KYC artefacts, ledger journal, and audit logs live exclusively on AWS me-central-1 (Abu Dhabi). DR in me-south-1 (Bahrain) is a CBUAE-approved alternate. Cross-border egress to vendors requires CBUAE notification and explicit customer consent in the consent register.

Encryption

TLS 1.3 in transit, AES-256-GCM at rest. Card PAN/CVV protected by a CloudHSM cluster with per-tenant data keys; root CMKs rotated quarterly. Field-level encryption for Emirates ID number, mobile, email.

Identity & access

Customer: OIDC via Auth0 with UAE Pass federation. Workforce: SSO + hardware-key MFA, time-bound break-glass procedures. Service-to-service: SPIFFE/SPIRE workload identities with rotating mTLS certs.

Detection & response

SIEM via Datadog Security; managed XDR through CrowdStrike Falcon. SOC operates 24×7 from M+9; on-call escalation paths for sev-1 incidents under 5-minute SLA.

Vulnerability mgmt

Snyk + GitHub Advanced Security on all repos; container scans in CI; quarterly external pen tests; annual red-team. CVSS 9+ remediated within 7 days, 7+ within 30.

Customer protections

Step-up auth (biometric or UAE Pass) for high-risk actions. Cooling-off period on new beneficiaries for the first 24 hours. Real-time alerts on every authorising payment.

Team

If we are going to build, we have to staff like a builder. The team is deliberately small, generalist by default, and AI-leveraged in every role. A pre-launch team of around thirty, growing to roughly sixty by year two and a hundred by year three. The shape matters more than the number. Each pod owns an outcome end to end. PMs ship features alongside engineers. Senior generalists outnumber narrow specialists, because what compounds in this kind of company is the ability to learn, not a particular skill.

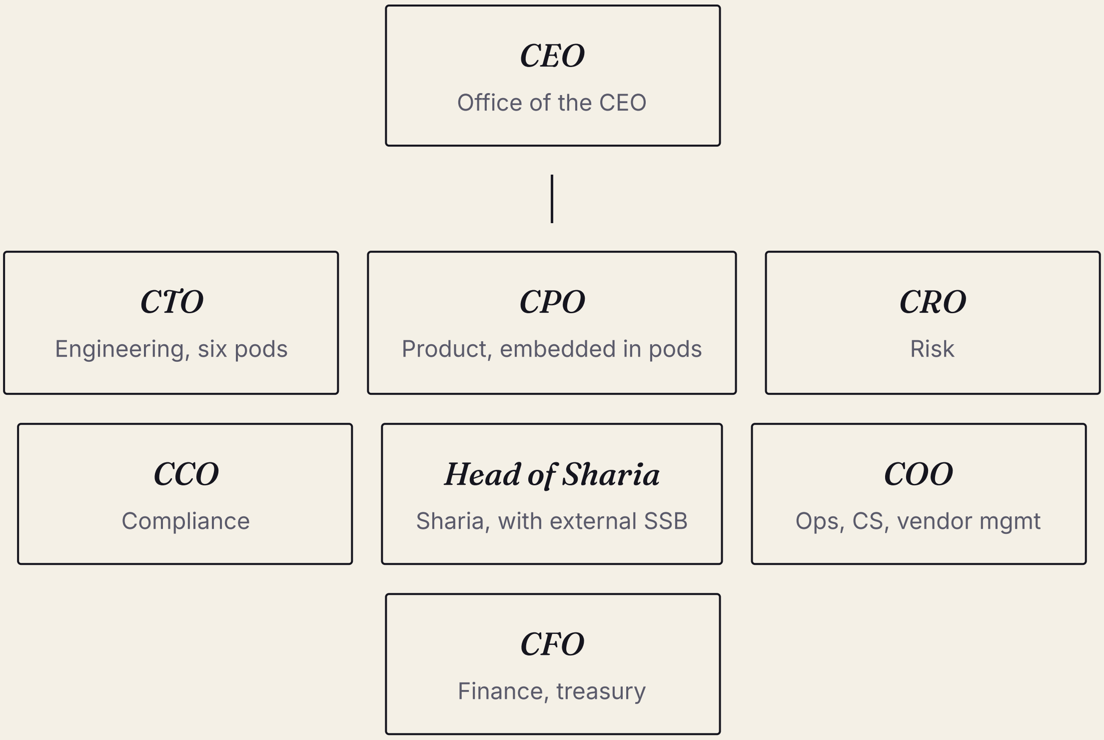
9.1 · The hiring philosophy

Three commitments shape who we hire and how the work flows.

- **General intelligence and agency over deep specialism.** Unless a role is specifically regulated (Sharia scholar, MLRO, CFO with banking experience), we hire generalists who can learn the domain quickly and who know how to use AI to multiply their output. Banking experience is useful, not required. What is required is the ability to think clearly, ship, and pick up a new system in a week.
- **Small pods, full ownership.** A pod is two to four people. It owns a customer outcome end to end (onboarding, decisioning, payments, accounting, expense, support). PMs are part of the pod and ship features alongside engineers. Design lives inside the pod. The pod sets its own cadence.
- **Minimal bureaucracy.** No multi-week sprint ceremonies. No long approval chains. Quarterly objectives, weekly check-ins, daily decisions made by the person closest to the problem.

9.2 · Year 2 shape

The Year 2 organisation is roughly six pods plus a thin spine of risk, compliance, finance, ops, and commercial. The shape:



Engineering and product (around 25)

Six pods of three to four. Each pod has a PM, two or three engineers, and a designer dropping in where needed. The pods own: customer apps and onboarding, decisioning and pricing, payments and Murabaha, accounts and FX, expense and accounting integrations, AI and data platform. SRE and security are part-time roles shared across pods, not separate teams. Senior engineers run the platform on the side.

Risk (around 6)

One CRO. Two credit officers who own the policy in the Decision Engine and review every Standard and Enhanced case. Two fraud ops who tune the fraud vendor and work the queues. One operational risk owner.

Compliance (around 6)

One CCO who is the regulator-facing leader. One MLRO. Two AML analysts. One KYC ops lead. One internal audit and DPO combined.

Sharia (2 internal plus 3 external)

One Head of Sharia who is full-time. One Sharia analyst who owns the product reviews and the audit sample. Three AAOIFI-certified scholars on retainer who meet monthly.

Operations and CS (around 12)

One COO. Bilingual CS team of six covering twelve hours a day. Three onboarding and recon ops. One vendor management lead. One workplace and BCP / DR. CS escalations into the engineering pods through a clear runbook, not a long handover.

Commercial (around 5)

One Head of Commercial. Two sales (SME field, broker / accountant channel). One partnerships lead (ERP partners, ASP partners). One brand and marketing generalist.

Finance (around 4)

One CFO. One treasury lead. One controller. One FP&A.

9.3 · Headcount over time

Function	*Year 1*	*Year 2*	*Year 3*
Engineering and product	10	25	40
Risk	3	6	10
Compliance	3	6	9
Sharia (internal)	1	2	3
Operations and CS	5	12	22
Commercial	3	5	9
Finance	2	4	6
G&A and CEO office	3	4	6
Total	**30**	**64**	**105**

These numbers assume an AI-leveraged team where each person produces what would have taken three to five people in a process-heavy bank. The traditional shape (twice the engineering, three times the ops) would deliver the same product later and at higher cost.

Bilingualism is non-negotiable

Every customer-facing role and every public-output role operates fluently in Arabic and English. The hiring filter reflects that.

Key architectural decisions

Each entry: the decision, the alternatives considered, and what we'd revisit it on.

ADR-001 *Mambu over Thought Machine for core banking*

Mambu's SaaS posture, UAE-region availability, mature deposit and loan products, and proven integrations with Sumsub, Lean, and Nymcard get us to MVP roughly six months faster than Thought Machine's smart-contract DSL approach. It is also Wio Bank's choice, which is a working reference at UAE production scale. We license at launch and migrate to an in-house replacement once we understand exact requirements from live operations.

Mambu (SaaS · UAE) | Thought Machine | 10x Banking | Finflux | Build in-house

Revisit on: volume > 50K active SMEs or a need for custom product types Mambu can't model.

ADR-001a *Decision Engine in-house, not vendor*

The decision engine is the platform's core IP. It composes deterministic policy gates, model scoring, and the offer engine. It must be configurable by the risk team day-to-day without engineering lift, ideally a low-code policy editor. Vendor scoring is generic. UAE SME risk is structurally different (high cashflow volatility, opaque ownership, trade-credit dependencies), and the data is our moat. We have a working prototype already.

Build in-house | Provenir | Experian DataX | SAS

ADR-002 *AWS me-central-1 over Azure UAE North*

AWS has the broader fintech ecosystem in the region, more managed services we'll actually use (MSK, Aurora Global, EKS, CloudHSM), and Mambu, Sumsub, and Lean all run on AWS in the region. CBUAE has approved both for production banking workloads.

AWS me-central-1 | Azure UAE North | OCI UAE

Revisit on: a strategic partner mandates Azure or a sovereign cloud requirement emerges.

ADR-003 *Event-sourced ledger via Kafka + Postgres projections*

The ledger is the heart of the bank. Event-sourcing gives auditability, replayability, and clean projections for reporting. Pure double-entry on Postgres alone would force us to invent the audit log; Kafka gives it for free.

Event-sourced · Kafka | Postgres-only double entry | Custom ledger on KSQL

Revisit on: Kafka operational burden outweighs benefits (unlikely on MSK).

ADR-004 *Sharia compliance modelled as contract type, not a downstream filter*

Every money-moving record carries its `contract_type` as a typed field, validated by a Sharia engine at write time. This is more invasive than a filter but removes whole classes of "compliance accidents" (booking interest as profit).

First-class contract type | Downstream filter | Manual review

ADR-005 *Build the underwriting model in-house*

Vendor scoring is generic. UAE SME credit risk is structurally different (high cashflow volatility, opaque ownership, trade-credit dependencies). The model is a core differentiator, and the data is our moat. The model class is whatever best fits the task at hand; the architecture commits to the platform that lets the team choose, not to one algorithm.

Build in-house | Provenir | Experian DataX

ADR-006 *TypeScript + Go split, not polyglot for its own sake*

TypeScript on every customer-facing service, Go on every payment- and money-touching service. Two stacks, two CI pipelines, one shared talent pool. We say no to Python on the request path.

ADR-007 *Nymcard over Marqeta*

Nymcard is MENA-native, integrates with UAE BIN sponsors out of the box, and pricing is friendlier at our launch volume. Marqeta is excellent in the US; for the UAE its overhead doesn't pay off until much higher scale.

Nymcard | Marqeta | Galileo

Revisit on: expansion to KSA + a need for unified processing across markets.

ADR-008 *Lean over building open banking in-house*

Building bank-by-bank connectors in the UAE is a 12-month project we'd never recoup. Lean has the UAE coverage; we own the orchestration on top.

Lean | Tarabut Gateway | Build

ADR-009 *Sumsub primary, FOCAL fallback for KYC/KYB*

Sumsub has the most complete UAE document coverage and the best UBO graph builder. FOCAL is a strong UAE-native backup with local support. Both behind a Mal-owned adapter; we can flip primary in minutes.

ADR-010 *Temporal for sagas, not Step Functions*

KYC, money movement, Murabaha lifecycle, dispute handling. All are multi-step, hours-to-days workflows with retries and compensation. Temporal's developer model (workflows as code) is dramatically more productive than Step Functions JSON.

ADR-011 *20-agent specialist swarm over a single all-purpose LLM*

Bounded specialist agents, each with a clear contract, tool list, and human-in-the-loop trigger, outperform a single monolithic LLM on every dimension that matters in a regulated lender. They are easier to evaluate (golden sets per agent), easier to cost-control (route the cheaper model for routine, the bigger model for memos and reasoning), easier to audit (per-agent trace logs), and safer (no agent can act outside its tool list). LangGraph on top of Temporal gives the orchestrator semantics. Bedrock gives UAE-region inference.

20 specialists + Temporal orchestrator | One general LLM | Buy off-shelf agent platform

ADR-012 *Dar Al Sharia as primary SSB*

Dar Al Sharia is the dominant UAE-resident Sharia advisory with deep AAOIFI-aligned practice and UAE Higher Sharia Authority recognition. Shariyah Review Bureau is the secondary choice (international footprint, strong Murabaha jurisprudence). Quarterly fatwa renewal + annual audit + 5% random sample of executed Murabaha contracts.

Dar Al Sharia | Shariyah Review Bureau | Mufti Taqi Usmani office

ADR-013 *ADGM Cat 2 + Cat 4 (with RegLab option)*

ADGM FSP gives us the credit licence (Cat 2) and the financing platform licence (Cat 4) in a single forum with a CBUAE-aligned regulator and an English-common-law jurisdiction. RegLab is available for a 12 to 18 month sandbox period for the BNPL instalment feature inside invoice finance, which is useful if we want a faster path to market for that single product.

ADGM FSP Cat 2 + Cat 4 | CBUAE Finance Company | DIFC

ADR-014 *Refinitiv + Dow Jones for AML, not ComplyAdvantage*

The UAE-specific watch-list coverage at Refinitiv (UAE national lists, CBUAE sanctions, Higher Committee Against Terrorism) and Dow Jones is materially better than the global-first ComplyAdvantage at our launch scale. Both behind a Mal-owned adapter; we can flip primary in minutes.

Refinitiv + Dow Jones | ComplyAdvantage | Acuris (Mergermarket)

Build over buy

The cost of building software has dropped dramatically. AI-assisted coding has automated something like 90 percent of routine implementation work, and the number is still climbing. What used to take a fifteen-person team six months can now be delivered by three to five high-agency developers and PMs who actually understand the business. The traditional SaaS licensing model, where you pay a vendor for a configurable but ultimately generic platform, is becoming less relevant for the parts of a bank that touch the customer, the policy, or the IP. The cost of building has collapsed. The cost of being locked into someone else's product roadmap has not.

11.1 · The exceptions

Regulator-mandated services

Credit bureau (AECB), licensed KYC and AML providers, sanctions screening, and commodity trading platforms for Sharia execution (DMCC, Bursa Suq Al-Sila). These are compliance requirements. They are not technology choices.

Fraud intelligence

Hub-and-spoke. A fraud vendor integrated across hundreds of institutions sees a compromised IP, a synthetic identity, or a spoofed device days before any individual lender can catch it internally. That network effect is not something you replicate by building. This is fundamentally different from a Decision Engine or an LOS, where the value is in your own proprietary logic.

11.2 · Licence at launch, migrate later

For systems where a mistake is costly on day one, we license to de-risk launch and migrate later. The clearest example is the LMS. Ledger accuracy is non-negotiable, and getting it wrong in production is not something you want to debug under pressure. We license a proven LMS (Mambu or equivalent) for launch, then build the in-house replacement once we understand exact requirements from live operations.

The architectural commitment that makes this work is composability. Every vendor-integrated module sits behind a clean API contract from day one, so the migration to an in-house module is a backend switch and not a re-architecture. We can run vendor and in-house side-by-side in shadow mode before cutting over. The same logic applies to collections, which starts manual at low volume and gets a dedicated platform as NPL volume justifies it.

11.3 · The lending stack: canonical component decisions

This is the table from the Vendor Selection note. It governs every component on the lending side.

Component	*Launch*	*Steady state*	*Why*
Decision Engine	Build in-house (Day 1)	In-house	Core IP. The risk team must be able to configure rules without engineering dependency. We have a working prototype.
LOS	Build in-house (Day 1)	In-house	Workflow is deeply product-specific. Off-the-shelf platforms force compromise on customer experience and funnel design.
Pricing Engine	Build in-house (Day 1)	In-house	Tightly coupled with the Decision Engine and the risk models. Must support real-time risk-based pricing, not static rate cards.
Collections Engine	Manual or light vendor	In-house	Low volume at launch, manual is fine. Build the dedicated platform as NPL volume scales. EWS is the priority investment (prevention over cure).
LMS	License (e.g. Mambu)	Migrate in-house	Ledger accuracy is non-negotiable. License a proven provider to de-risk launch, then migrate once we know exact requirements.
Core Banking	Integrate existing	Integrate existing	Depends on Mal's existing core banking stack. Standard integration layer.
Fraud Management	Buy (vendor)	Buy (vendor)	Hub-and-spoke. Vendor aggregates signals across the industry. Network effect is irreplaceable.
Bureau / KYC / AML	Buy / API	Buy / API	Regulatory requirement. Standard API integration with AECB, licensed KYC providers, sanctions databases.
Commodity Trading (Islamic)	Buy / partner	Buy / partner	DMCC or Bursa Suq Al-Sila for Tawarruq commodity execution. Regulated marketplace, not a build candidate.
Portfolio Monitoring & Reporting	Build in-house	In-house	Real-time dashboarding and EWS are competitive differentiators. Must be tailored to our risk framework and KPI hierarchy.

11.4 · Extending the same logic to the SME banking surface

The lending stance carries over to the rest of the bank. The same questions apply. Does this touch our customers, our policy, or our IP? Then build it. Is it regulator-mandated or a hub-and-spoke commodity? Then buy it. Is the launch risk too high to absorb? Then license now and migrate later.

SME banking component	*Launch*	*Steady state*	*Why*
Customer apps (iOS, Android, web, ops console)	Build	Build	This is the product. A small senior team ships faster on its own stack than any vendor allows. Distinctive experience is the brand.
Onboarding orchestration (KYB workflow)	Build	Build	Workflow IP. Sequencing the UAE Pass, DET / DED / ADGM / DIFC, AECB, MOET UBO, Lean, and Peppol calls is a competitive moat. Vendor BPM tools cannot match.
Customer graph (KYB record, UBO graph, signatories, consent register)	Build	Build	The canonical record of who our customers are. Lives at the centre of every other system. Never outsourced.
Identity verification (face match, liveness, document forensics)	Buy (Sumsub, FOCAL)	Buy	Specialised computer vision and a network of fraud signals. Building this in-house is the wrong battle for a lender.
UAE Pass federation	Integrate	Integrate	Government identity rail. Standard integration.
Multi-currency accounts (customer-facing wallet experience)	Build	Build	The customer's view of their money. We build the experience; the FX counterparty sits behind it.
FX engine (rates, hedging, sub-ledger for non-AED)	Buy (Currencycloud)	Buy / hybrid	Liquidity, hedging, regulatory licensing for FX. Commodity infrastructure. Build a hedging policy on top; do not build the FX book.
Virtual IBAN issuance	Partner (Wio, RAKBank)	Self-issue once licensed	Today we lean on a partner bank's IBAN. Once Mal holds the licence, self-issue.
Card issuance and processing	Buy (Nymcard)	Buy	BIN sponsorship, scheme connectivity, 3DS, tokenisation. Commodity rails.
Expense management (per-employee cards, MCC policy, receipt OCR, approvals)	Build	Build	Customer-facing surface. The card rail is bought; the policy engine, receipt OCR, approval workflows, and reporting are ours.

SME banking component	*Launch*	*Steady state*	*Why*
Accounting connectors (Wafeq, Zoho, QuickBooks, Xero, Tally)	Build (native)	Build (native)	The native experience lives in our app. Codat is the meta-integrator for the long tail only. The five core connectors are ours.
Open banking aggregator	Buy (Lean, first CBUAE IPA)	Buy	Bank-by-bank integration is a 12-month job we would never recoup. Lean owns the breadth.
Payments orchestration (rail selection, idempotency, retries, sagas)	Build	Build	Money movement IP. Each rail (AANI, UAEFTS, IPP, Buna, SWIFT) is integrated; orchestration logic is ours.
Payment rails (AANI, UAEFTS, IPP, Buna, SWIFT)	Integrate	Integrate	National infrastructure. Standard integration through partner bank where required.
Peppol e-invoicing	Buy ASP (Avalara + RTC Suite)	Buy ASP	FTA-accredited service provider is required. We pick two for resilience and own the integration layer.
Sanctions, PEP, adverse media	Buy (Refinitiv, Dow Jones)	Buy	Regulator-mandated screening with continuous list updates. Vendor required.
Document OCR (Arabic-English VAT invoices, IDs, MOA)	Buy + fine-tune	Buy + fine-tune	Vendor OCR for the substrate; we fine-tune layout and field-extraction on our own corpus. Best of both.
Murabaha contract engine	Build	Build	Template-driven, SSB-approved. Refuses to render a contract outside template. Core Sharia IP.
Sharia screening reference data	Buy (IdealRatings)	Buy	Industry classifications for halal compatibility. Specialist reference data, not worth replicating.
Conversational AI (Arabic-first CX)	Build on managed models	Build	The customer-facing assistant is ours. We use managed LLM serving (Bedrock for Claude, fine-tuned Cohere or open-weights for Arabic) but build the RAG, the policy grounding, and the eval harness in-house.

SME banking component	*Launch*	*Steady state*	*Why*
Foundation LLM serving	Buy (AWS Bedrock)	Buy	Cloud-resident inference in UAE region. Commodity. We pay for the inference, not the platform.
Feature store, embedding store, vector retrieval	Build on managed primitives	Build	The platform that lets the team choose, version, and replace models. Built on managed databases and managed vector stores.
Observability (infra, app traces)	Buy (Datadog)	Buy	Commodity. Buy.
Observability (LLM traces, evals)	Buy (Langfuse self-hosted)	Buy	Specialist tool, self-hosted for data residency. Commodity.
Security (EDR, SIEM, SOC tooling)	Buy	Buy	Vendor benefit from cross-customer telemetry. Same network-effect logic as fraud.
Identity provider (workforce SSO + MFA)	Buy (Auth0, hardware keys)	Buy	Commodity.
Workflow / sagas / agent orchestration	Buy (Temporal Cloud)	Buy	Durability, retries, replay are well-served by a managed product.
Portfolio monitoring, EWS, regulatory reporting	Build	Build	Real-time, tailored to our risk framework. The whole point is that it is ours.
Cloud (compute, storage, network)	Buy (AWS me-central-1)	Buy	Commodity. UAE-resident, CBUAE-approved.

11.5 · The culture this requires

Build-first only works with a specific kind of team. Three commitments make it possible.

PMs ship features alongside engineers

The old chain of PM, design, engineering, QA, release is too slow for this world. When AI handles most of the implementation, the PM's job shifts from writing specs to actually shipping. We set this up deliberately.

Small teams, high agency

Five to ten people with high agency and domain knowledge will outperform a fifty-person team that runs on process. AI-assisted development means the bottleneck is no longer coding capacity. It is clarity of thought about what to build.

Minimal bureaucracy

Approval chains and multi-week sprint ceremonies kill velocity. The companies winning right now have stripped out unnecessary process and given individual contributors real ownership. Small teams. Daily decisions made by the person closest to the problem.

A year ago, the recommendation would have been heavier on vendor-assisted launches across the board. What is observable in the market right now, with how quickly AI-assisted coding is maturing, is that building in-house is the better long-term bet. The SaaS licensing model served the industry well when software required large teams and long timelines. That is changing. The organisations that recognise the shift earliest will have a structural advantage that compounds.

Indicative steady-state vendor spend at Year 1 scale (~2,500 active SMEs, ~250K monthly transactions, ~AED 500M in financed receivables) is AED 4 to 6M across cloud, identity, screening, bureau, card rails, OCR, ASP, and managed model serving. Most contracts step up by roughly 50 percent in Year 2 and 2x in Year 3 as volume grows. The number to watch is not vendor cost, but the cost of running engineering with the wrong-sized team. A 50-person stack that drifts toward process is more expensive than a 10-person stack that ships.

Roadmap and cost

A 24-month plan tied to regulatory milestones. Each phase has a clear technical exit criterion and a cost band.

Q3 2026 · Foundations (launch quarter)

- ADGM FSP Cat 2 + Cat 4 licence approved; SSB (Dar Al Sharia) onboarded; founding team in place.
- Mambu live; AWS me-central-1 landing zone; IaC baseline (Terraform + AWS Control Tower).
- UAE Pass federation live; AECB Etihad Bureau API connected; Sumsub + Refinitiv + Dow Jones live in staging.
- Murabaha contract template approved by SSB + submitted to UAE Higher Sharia Authority.
- First four AI agents shipped: KYB Orchestrator, Document Parser, Sanctions Screener, Customer Support Tier 1.

Exit: happy-path Murabaha invoice finance demoable in staging; ledger journal end-to-end correct; no real money moved.

Q4 2026 · Closed beta

- First 200–500 beta SMEs onboarded via private invite from operating-account base.
- Multi-currency wallets (AED + USD) live; virtual cards via Nymcard; Aani + UAEFTS rails connected.
- Supplier-side BNPL 30/60/90 live; Open Finance direct-debit mandates live via Lean.
- Accounting integrations: Wafeq + Zoho Books. Peppol ASP integration (Avalara + RTC Suite).
- Next four agents: Bank Statement Analyst, Reconciliation Agent, Anomaly Detector, Knowledge Base Curator.

Exit: 500 SMEs transacting; AED 50M financed cumulatively; < 1 incident per 1,000 transactions; weekly cohort review with CRO.

Q1 2027 · Public launch + instalment plan

- Open enrolment opens. Auto-underwriting (Express) at ticket ≤ AED 250K; manual above.
- Instalment plan (2 to 6 months) launches inside invoice finance. This is the headline buyer-side differentiator versus Comfi, Beehive, Funding Souq.
- Sharia Murabaha contract live for every customer (single SKU).
- Expense management surfaces ship: receipt OCR, MCC controls, per-employee cards, policy engine.
- Underwriting + collections agents: Credit Memo Writer, Invoice Verifier, Collections Conductor.

Exit: 2,500 SMEs onboarded; AED 250M financed in the quarter; median Express decision < 4 hours.

Q2 2027 · Scale + AI deepening

- Confidential mode for invoice finance (buyer sees supplier invoice with Mal as payment destination; supplier paid Day 1).
- Supplier non-recourse pricing live (up to 15% of book in Y1).
- Risk + Sharia agents: Settlement Negotiator, Early Warning Agent, AML Transaction Monitor.
- AI-driven cash-flow forecasting live in customer dashboards.
- CBUAE Customer Protection Regulation compliance lock-down (effective 13 Sep 2026 → must be in production by Q2 2027).

Exit: 5,000 SMEs, AED 500M financed in the quarter, NPS > 60.

Q3–Q4 2027 · Integrated accounting + cross-border

- Embedded Wafeq + Zoho + QuickBooks + Xero + Tally integrations live; Codat for the long tail.
- Cross-border via Buna (AED corridor) + SWIFT (USD/EUR/GBP) live.
- Conversational AI assistant graduates to private preview, then GA.
- Remaining agents: Pricing Optimizer, Cohort Analyst, Regulatory Horizon Scanner.
- SME Peppol mandate goes live (July 2027). Mal is already production-ready.

Exit: 12,000 SMEs, AED 1.2B financed cumulatively in Y2, ops cost per active SME < AED 60/month.

Cost, steady state Year 2

People (around 60 FTE) **~AED 65M** small senior team, AI-leveraged	*Vendors and infra* **~AED 8M** cloud, SaaS, rails, ASPs
Regulator, audit, legal **~AED 5M** ADGM, CBUAE, external audit, SSB	*GTM* **~AED 10M** marketing, sales, partnerships

Total Year 2 opex is roughly AED 90M, almost half what a traditionally-staffed bank at the same volume would carry. The lower figure is the consequence of the build-first, small-team approach.

> *What this architecture is built to survive*
>
> *Adding a new payment rail (a quarter of work, one adapter). Switching KYC vendor (two weeks, no domain code changes). Onboarding a new Sharia product type (template approval + engine config, no infrastructure changes). Surviving an AZ failure (automatic, < 1 minute). Surviving a full regional failure (DR plan, < 15 minutes). A CBUAE audit (read-only console, audit log, signed receipts).*

Mal · capital that moves with you.

Architecture documentation · v1.0 · 2026-05-13 · Companion to the visual architecture diagram

Prepared for the executive team and the board. Distribute internally only.